



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 6

- 1. Case Study Title:** To demonstrate JavaScript string operations using `substring()`, `indexOf()`, `split()`, `replace()`, string reversal, and vowel counting based on user input.

Case Study Program Code:

```
<!  
function performOperations() {  
  
    let str = document.getElementById("str").value;  
  
    let start = Number(document.getElementById("start").value);  
    let end = Number(document.getElementById("end").value);  
  
    let search = document.getElementById("search").value;  
    let separator = document.getElementById("separator").value;  
  
    let oldText = document.getElementById("oldText").value;  
    let newText = document.getElementById("newText").value;  
  
    let output = "";  
  
    output += "<div class='result-title'>Results</div>";  
  
    // substring()  
    output += "Substring: " +  
        str.substring(start, end) + "<br>";  
  
    // indexOf()  
    output += "Index of '" + search + "': " +  
        str.indexOf(search) + "<br>";  
  
    // split()  
    if (separator.toLowerCase() == "space") {  
        output += "Split: " +  
            str.split(" ") + "<br>";  
    }  
}
```

</head>

<body>

<h2>

</h2>



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```
<label>Substring Start:</label>
<input type="number" id="start" placeholder="Enter start index">
<br>
<label>Substring End:</label>
<input type="number" id="end" placeholder="Enter end index">
<br>
<label>IndexOf - Enter text to search:</label>
<input type="text" id="search" placeholder="Enter character or
word">
<br>
<label>Split - Enter separator:</label>
<input type="text" id="separator" placeholder="Example: space">
<br>
<label>Replace - Text to replace:</label>
<input type="text" id="oldText" placeholder="Enter text to replace">
<br>
<label>Replace - New text:</label>
<input type="text" id="newText" placeholder="Enter new text">
<br>
<button onclick="performOperations()">Perform Operations</button>
<div id="output"></div>
<script src="shreya_case6.js"></script>
```

</body>

</html>

```
function performOperations() {

    let str = document.getElementById("str").value;

    let start = Number(document.getElementById("start").value);
    let end = Number(document.getElementById("end").value);

    let search = document.getElementById("search").value;
    let separator = document.getElementById("separator").value;
```



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```
let oldText = document.getElementById("oldText").value;
let newText = document.getElementById("newText").value;
let output = "";
// Substring
output += "Substring: " + str.substring(start, end) + "<br>";
// IndexOf
output += "Index of '" + search + "': " + str.indexOf(search) +
"<br>";
// Split
if (separator === "space") {
    output += "Split: " + str.split(" ") + "<br>";
} else {
    output += "Split: " + str.split(separator) + "<br>";
}
// Replace
output += "Replace: " + str.replace(oldText, newText) + "<br>";

// Reverse
let reverse = str.split("").reverse().join("");
output += "Reversed String: " + reverse + "<br>";
// Count Vowels
let count = 0;
for (let i = 0; i < str.length; i++) {
    let ch = str[i].toLowerCase();
    if (ch == "a" || ch == "e" || ch == "i" ||
        ch == "o" || ch == "u") {
        count++;
    }
}
output += "Number of Vowels: " + count;
document.getElementById("output").innerHTML = output;
}
```



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3. Output:

The screenshot shows a web browser window titled "JavaScript String Operations". The address bar shows the file path: "C:/Users/HP/Desktop/24070521207_SharvaniW(US)/casestudy6.html". The main content area has a yellow background. In the center, there is a white box titled "String Operations" with the subtitle "Perform different string operations using JavaScript". Below this, there is a form with the following fields:

- Enter String / Paragraph:** A text area containing the text: "Mirror Mirror on the wall, I always get up after i fall, whether i run walk or crawl , i set my goals and achieve them all".
- Substring Start:** A text input field with the value "5".
- Substring End:** A text input field with the value "11".
- Character / Word to Find:** A text input field with the value "achieve".
- Split Separator:** A text input field with the value "space".
- Text to Replace:** A text input field with the value "Goals".
- New Text:** A text input field with the value "Aims".

The screenshot shows the same web browser window, but now the "Perform Operations" button has been clicked, and the results are displayed in a white box below the input fields. The results are as follows:

- Substring:** r Mirr
- Index of 'achieve':** 106
- Split:** Mirror,Mirror,on,the,wall,,I,always,get,up,after,i,fall,,whether,i,run,walk,or,crawl,,,i,set,my,goals,and,achieve,them,all
- Replace:** Mirror Mirror on the wall, I always get up after i fall, whether i run walk or crawl , i set my goals and achieve them all
- Reversed String:** lla meht eveihca dna slaog ym tes i , lwarc ro klaw nur i rehtehw ,llaf i retfa pu teg syawla I ,llaw eht no rorriM rorriM
- Number of Vowels:** 34

4. Result/Conclusion:

Thus, various JavaScript string operations were successfully performed, including **substring extraction**, **text searching**, **splitting**, **replacing**, **reversing**, and **vowel counting**, and the results were displayed dynamically on the webpage.



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● Experiment No.: 1

- 1. Title:** Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.
- 2. Software/Tools Required:** Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.
- 3. Theory:** JavaScript is a scripting language used to make web pages interactive. It can be added in three ways:
 - Inline JavaScript: Written inside HTML elements.
 - Internal JavaScript: Written inside the `<script>` tag in the HTML file.
 - External JavaScript: Written in a separate `.js` file and linked to the HTML page.
 - Console methods like `console.log()`, `console.info()`, `console.warn()`, and `console.error()` are used for debugging and displaying messages.



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4. Output:

file:///C:/Users/HP/Desktop/sharvariw(1207)index.html

MySpace

Hello, I'm Sharvari

Welcome to my little corner of the internet. I love technology, creativity, and building amazing things. I am a passionate learner with a curiosity for technology, creativity, and innovation. I enjoy designing clean, modern websites, solving real-world problems through code, and continuously exploring new skills. This website gives you a glimpse into my journey, my interests, and the work I'm excited to create.

[Know More](#)

C:/Users/HP/Desktop/myportfolio/index.html#about

MySpace

[Home](#) [About](#)

My Skills

HTML

CSS

JavaScript

Python

Java

Problem Solving

Contact

Email : sharvari@gmail.com

LinkedIn : [linkedin.com/in/sharvari](https://www.linkedin.com/in/sharvari)

[Change Footer](#)

[Show Contact](#)

[Show Date & Time](#)

5. Result/Conclusion: The experiment was performed successfully. A webpage displaying a welcome message and user information was created using inline, internal, and external JavaScript, and console methods were demonstrated.



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Case Study No.: 1

1. **Case Study Title:** Create a webpage for college department and link the student information page with it and write a java script to display “welcome to sit nagpur” in an alert box.

2. Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR</title>
  <style>
    body {
      font-family: 'Times New Roman', Times, serif;
      background-color: #FFB6C1;
      text-align: center;
      margin: 0;
      padding: 0;
    }
    h1 {
      color: brown;
    }
    .welcome-msg {
      font-size: 1.5rem;
      color: #333;
      margin: 20px 0;
      font-weight: bold;
    }
    .form-container {
      margin-top: 20px;
    }
    input {
      margin: 5px;
      padding: 8px;
      width: 250px;
    }
    button {
      padding: 8px 15px;
      background-color: brown;
      color: white;
      border: none;
      cursor: pointer;
      font-weight: bold;
    }
    button:hover {
      background-color: #6e1a1a;
    }
  </style>
</head>
<body>
  <h1>SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR</h1>
  <div class="welcome-msg">
    <p>WELCOME TO SIT NAGPUR</p>
  </div>
  <div class="form-container">
    <input type="text" value="Student Information Page" />
    <button type="button" value="Link" />
  </div>
</body>
</html>

/* --- CUSTOM POPUP MODAL STYLES --- */
```



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```
.modal-overlay {
  position: fixed;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background-color: rgba(0, 0, 0, 0.5); /* Dims the background */
  display: flex;
  justify-content: center;
  align-items: center;
  z-index: 1000;
}

.modal-box {
  background-color: white;
  padding: 30px;
  border-radius: 8px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.2);
  text-align: center;
  max-width: 400px;
  width: 90%;
}

.modal-box h4 {
  margin-top: 0;
  color: brown;
  font-size: 1.3rem;
}

.modal-buttons {
  margin-top: 20px;
  display: flex;
  justify-content: center;
  gap: 15px;
}

.btn-yes {
  background-color: #28a745;
}

.btn-no {
  background-color: #dc3545;
}
</style>
</head>
<body>

<!-- CUSTOM POPUP MODAL -->
<div id="customModal" class="modal-overlay">
  <div class="modal-box">
    <h4>Registration Check</h4>
    <p id="modalQuestion">Are you already registered with us?</p>
    <div class="modal-buttons" id="actionButtons">
      <!-- User clicks YES -->
      <button class="btn-yes" onclick="handleUserChoice(true)">Yes, I am</button>
      <!-- User clicks NO -->
      <button class="btn-no" onclick="handleUserChoice(false)">No, I'm new</button>
    </div>
  </div>
</div>
</div>
```



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<!-- MAIN BACKGROUND CONTENT -->

<header>

<h1>SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR</h1>

<p class="welcome-msg">Welcome to the Student Portal!</p>

<hr>

<h2>Department Of Computer Science and Engineering</h2>

<h3>Specialization: AI & ML / Data Science</h3>

</header>

<main class="form-container">

<h3>Student Information Form</h3>

<input type="text" id="nameInput" placeholder="Enter Full Name">

<input type="text" id="rollInput" placeholder="Enter Roll Number">

<input type="email" id="emailInput" placeholder="Enter Email Address">

<button onclick="collectData()">Submit Data</button>

</main>

<script>

// Handles what happens after clicking Yes or No

function handleUserChoice(isRegistered) {

const questionText = document.getElementById("modalQuestion");

const buttonsContainer = document.getElementById("actionButtons");

// 1. Change the text to display the "Welcome to SIT" message

if (isRegistered) {

questionText.innerHTML = "Welcome to SIT!
Please proceed to login.";

} else {

questionText.innerHTML = "Welcome to SIT!
Please create an account on our registration page.";

}

// 2. Replace the Yes/No buttons with a single "OK / Close" button to avoid browser blocks

buttonsContainer.innerHTML = '<button onclick="closeModal()">OK</button>';

}

// Closes the modal box completely to reveal the form

function closeModal() {

document.getElementById("customModal").style.display = "none";

}

// Logic for collecting form data

function collectData() {

const name = document.getElementById("nameInput").value;

const rollNo = document.getElementById("rollInput").value;

const email = document.getElementById("emailInput").value;

if (!name || !rollNo || !email) {

alert("Please fill out all fields.");

return;

}

localStorage.setItem("studentName", name);

localStorage.setItem("studentRoll", rollNo);

localStorage.setItem("studentEmail", email);

window.location.href = "student.html";

}



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```
</script>
</body>
</html>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Information - SIT</title>
  <style>
    body {
      font-family: 'Times New Roman', Times, serif;
      background-color: #FFB6C1;
      text-align: center;
      padding: 20px;
    }
    h1 {
      color: brown;
    }
    .info-card {
      background-color: white;
      max-width: 400px;
      margin: 30px auto;
      padding: 20px;
      border-radius: 8px;
      box-shadow: 0px 4px 8px rgba(0,0,0,0.1);
      text-align: left;
    }
    .info-item {
      margin-bottom: 15px;
      font-size: 1.1rem;
    }
    .label {
      font-weight: bold;
      color: brown;
    }
    a {
      display: inline-block;
      margin-top: 20px;
      color: brown;
      text-decoration: none;
      font-weight: bold;
    }
  </style>
</head>
<body>

  <h1>SIT Student Profile</h1>
  <h2>Submitted Information</h2>

  <div class="info-card">
    <div class="info-item"><span class="label">Name:</span> <span id="displayName"></span></div>
    <div class="info-item"><span class="label">Roll Number:</span> <span id="displayRoll"></span></div>
    <div class="info-item"><span class="label">Email:</span> <span id="displayEmail"></span></div>
  </div>
```



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[< Go Back to Form](index.html)

<script>

// Fetch the stored data on page load

window.onload = function() {

const name = localStorage.getItem("studentName");

const rollNo = localStorage.getItem("studentRoll");

const email = localStorage.getItem("studentEmail");

// Display data, or show a fallback message if empty

document.getElementById("displayName").innerText = name ? name : "No data found";

document.getElementById("displayRoll").innerText = rollNo ? rollNo : "No data found";

document.getElementById("displayEmail").innerText = email ? email : "No data found";

}

</script>

</body>



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Output:

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/casestudy1index.html

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Welcome to the Student Portal!

Department Of Computer Science and Engineering

Specialization: AI & ML / Data Science

Student Information Form

Registration Check

Are you already registered with us?

Yes, I am

No, I'm new

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/casestudy1index.html

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Welcome to the Student Portal!

Department Of Computer Science and Engineering

Specialization: AI & ML / Data Science

Student Information Form

Robert

1207

robertiscool@gmail.com

Submit Data



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C:/Users/HP/Desktop/24070521207_SharvariW(JS)/casestudy1student.html

SIT Student Profile

Submitted Information

Name: robert

Roll Number: 1207

Email: robertiscool@gmail.com

[← Go Back to Form](#)

3. **Result/Conclusion:** The college department webpage was created successfully. The Student Information page was linked to it, and JavaScript was implemented using inline, internal, and external methods. A welcome message was displayed in an alert box, and `console.log()` output was verified in the browser console.



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● Experiment No.: 2

1. Title: Use `var`, `let`, `const`, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: `var` is function-scoped, `let` is block-scoped, and `const` is used for fixed values.

The `prompt()` function accepts user input, and `Number()` converts input into numeric values.

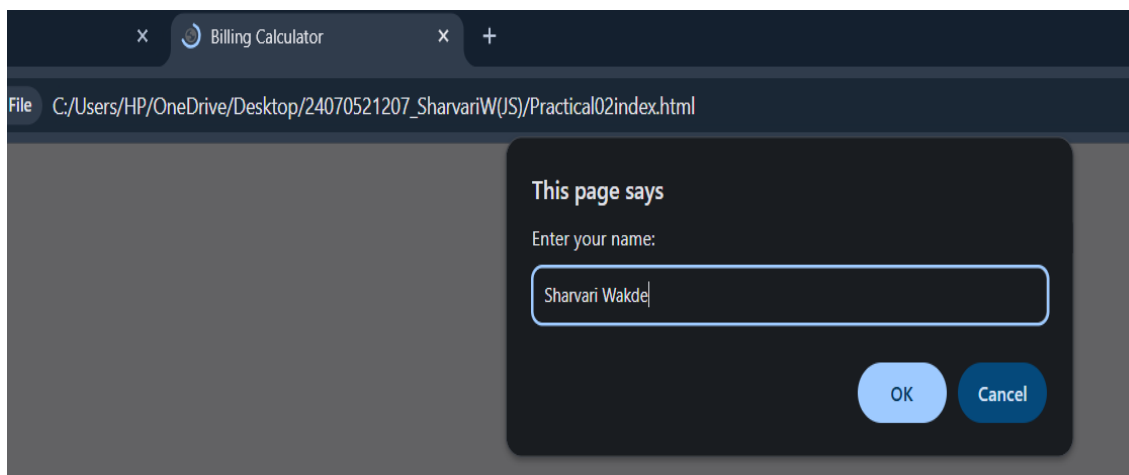
The program calculates the total amount, GST, and final bill using arithmetic operations.

An object stores all billing details together, while object destructuring extracts required values easily.

Template literals (``) are used to display formatted output with variable values using `$ { }`.

The `document.write()` method displays the billing receipt on the webpage.

4. Output :





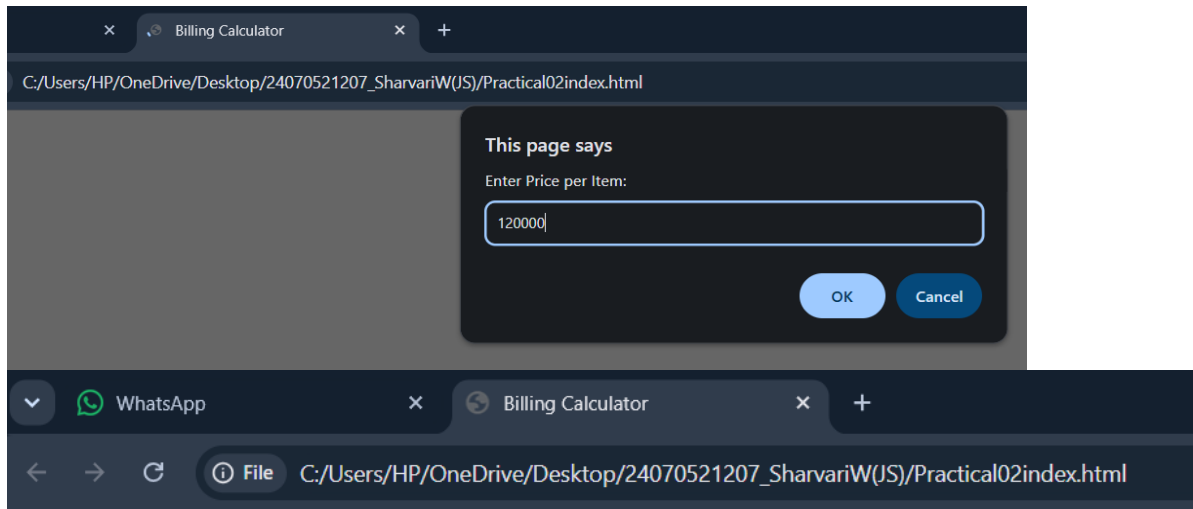
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Billing Calculator

Billing Receipt

Customer: Sharvari Wakde

Product: laptop

Quantity: 3

Price per Item: 120000

Total: 360000

GST (18%): 64800.00

Final Bill: 424800.00

5. **Result/Conclusion:** A billing calculator was successfully created using JavaScript. The program accepted user input, calculated the bill with GST, and demonstrated the use of var, let, const, template literals, and object destructuring effectively.



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Case Study No.: 2

- Case Study Title:** Develop the webpage for an online shopping website with using html,css and html tags and calculate bill,customer name,mobile no,invoice no,product no,quantity(kg),rate per kg,discount,GST,packing charges,payment mode(cash/card/UPI), Membership(yes/no).

2. Case Study Program Code:

```
<!DOCTYPE html>

<html>

<head>

<title>Online Shopping Billing System</title>


<style>
    body{
        font-family: Arial, sans-serif;
        background:#f2f2f2;
    }

    .container{
        width:600px;
        margin:30px auto;
        background:white;
        padding:20px;
        border-radius:10px;
        box-shadow:0px 0px 10px gray;
    }

    h2{
        text-align:center;
        color:#0066cc;
    }

    table{
        width:100%;
    }

    td{
        padding:8px;
```



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}

```
input,select{
  width:100%;
  padding:8px;
}
```

```
button{
  width:100%;
  padding:10px;
  background:#0066cc;
  color:white;
  border:none;
  font-size:16px;
  cursor:pointer;
  border-radius:5px;
}
```

```
button:hover{
  background:#004c99;
}
```

```
#result{
  margin-top:20px;
  background:#e8f5e9;
  padding:15px;
  border-radius:5px;
  font-size:18px;
}
```

</style>

</head>

<body>

<div class="container">

<h2>ONLINE SHOPPING BILL</h2>

<table>

<tr>

<td>Customer Name</td>



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<td><input type="text" id="cname"></td>

</tr>

<tr>

<td>Mobile Number</td>

<td><input type="text" id="mobile"></td>

</tr>

<tr>

<td>Invoice Number</td>

<td><input type="text" id="invoice"></td>

</tr>

<tr>

<td>Product Name</td>

<td><input type="text" id="product"></td>

</tr>

<tr>

<td>Quantity (Kg)</td>

<td><input type="number" id="qty"></td>

</tr>

<tr>

<td>Rate per Kg</td>

<td><input type="number" id="rate"></td>

</tr>

<tr>

<td>Discount (%)</td>

<td><input type="number" id="discount"></td>

</tr>

<tr>

<td>GST (%)</td>

<td><input type="number" id="gst"></td>

</tr>

<tr>

<td>Packing Charges</td>

<td><input type="number" id="packing"></td>

</tr>



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```
<tr>
<td>Payment Mode</td>
<td>
<select id="payment">
<option>Cash</option>
<option>Card</option>
<option>UPI</option>
</select>
</td>
</tr>

<tr>
<td>Membership</td>
<td>
<select id="member">
<option>Yes</option>
<option>No</option>
</select>
</td>
</tr>

<tr>
<td colspan="2">
<button onclick="calculateBill()">Generate Bill</button>
</td>
</tr>

</table>

<div id="result"></div>

</div>

<script>

function calculateBill()
{

var cname=document.getElementById("cname").value;
var mobile=document.getElementById("mobile").value;
var invoice=document.getElementById("invoice").value;
var product=document.getElementById("product").value;
```



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```
var qty=parseFloat(document.getElementById("qty").value);
var rate=parseFloat(document.getElementById("rate").value);
var discount=parseFloat(document.getElementById("discount").value);
var gst=parseFloat(document.getElementById("gst").value);
var packing=parseFloat(document.getElementById("packing").value);
```

```
var payment=document.getElementById("payment").value;
var member=document.getElementById("member").value;
```

```
var amount=qty*rate;
```

```
var discountAmount=(amount*discount)/100;
```

```
var subtotal=amount-discountAmount;
```

```
var gstAmount=(subtotal*gst)/100;
```

```
var total=subtotal+gstAmount+packing;
```

```
if(member=="Yes")
{
    total=total-(total*5/100);
}
```

```
document.getElementById("result").innerHTML=
```

```
"<h3>Invoice</h3>"+
```

```
"<b>Customer Name :</b> "+cname+"<br><br>"+
```

```
"<b>Mobile No :</b> "+mobile+"<br><br>"+
```

```
"<b>Invoice No :</b> "+invoice+"<br><br>"+
```

```
"<b>Product :</b> "+product+"<br><br>"+
```

```
"<b>Quantity :</b> "+qty+" Kg<br><br>"+
```

```
"<b>Rate :</b> ₹"+rate+" /Kg<br><br>"+
```

```
"<b>Amount :</b> ₹"+amount.toFixed(2)+"<br><br>"+
```

```
"<b>Discount :</b> ₹"+discountAmount.toFixed(2)+"<br><br>"+
```



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```
"<b>GST :</b> ₹"+gstAmount.toFixed(2)+"<br><br>"+  
  
"<b>Packing Charges :</b> ₹"+packing.toFixed(2)+"<br><br>"+  
  
"<b>Payment Mode :</b> "+payment+"<br><br>"+  
  
"<b>Membership :</b> "+member+"<br><br>"+  
  
"<h2>Total Bill : ₹"+total.toFixed(2)+"</h2>";  
  
}  
  
</script>  
  
</body>  
</htm
```

Output:

file:///C:/Users/HP/Desktop/sharvariw(1207)/casestudy2.html

ONLINE SHOPPING BILL	
Customer Name	Sharvari
Mobile Number	456789765
Invoice Number	23000
Product Name	mobile
Quantity (Kg)	1
Rate per Kg	230988
Discount (%)	14
GST (%)	18
Packing Charges	200
Payment Mode	Cash
Membership	Yes
Generate Bill	



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file:///C:/Users/HP/Desktop/sharvariw(1207)/casestudy2.html

Invoice

Customer Name : Sharvari

Mobile No : 456789765

Invoice No : 23000

Product : mobile

Quantity : 1 Kg

Rate : ₹230988 /Kg

Amount : ₹230988.00

Discount : ₹32338.32

GST : ₹35756.94

Packing Charges : ₹200.00

Payment Mode : Cash

Membership : Yes

Total Bill : ₹222876.29

Result/Conclusion: The online shopping billing webpage was successfully developed using **HTML, CSS, and JavaScript**. The program accepted customer and product details, calculated the total bill including **discount, GST, and packing charges**, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.



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● Experiment No.: 3

1. **Title:** Implement control structures and form validation; create a grading system based on user-entered marks.

2. **Software/Tools Required:** VS Code, web browser, HTML, JavaScript.

3. **Theory: Control structures** (if, else if, and else) are used to make decisions based on the marks entered by the user.

Form validation checks whether the input is empty or outside the valid range (0–100) before processing.

The `getElementById()` method is used to access the value entered in the input box.

The `Number()` function converts the entered value from text to a numeric value. A **function** (`calculateGrade()`) is called when the **Submit** button is clicked to validate the input and calculate the grade.

The calculated grade is displayed on the webpage using the `innerHTML` property.

4. Output :

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

 **Sharvari's Grading System**

Enter Marks:

Marks: 95
Grade: A Grade



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C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

This page says

Please enter marks!

OK

 **Sharvari's Grading System**

Enter Marks:

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

This page says

Please enter valid marks between 0 and 100.

OK

 **Sharvari's Grading System**

Enter Marks:

5. Result/Conclusion:The grading system was successfully implemented using JavaScript. The program validated the user input, assigned grades using if-else control structures, and displayed the correct grade based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



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Case Study No.: 3

- 1. Case Study Title:** Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

Write a JavaScript program that accepts student marks and display the Scholarship category.

Case Study Program Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Scholarship Eligibility</title>
```

```
<style>
```

```
body {
```

```
margin: 0;
```

```
font-family: Arial, sans-serif;
```

```
background: linear-gradient(135deg, hsl(21, 75%, 69%), hsl(39, 52%, 79%));
```

```
display: flex;
```

```
justify-content: center;
```

```
align-items: center;
```

```
height: 100vh;
```



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}

.container {

background: white;

padding: 35px;

border-radius: 15px;

box-shadow: 0 8px 20px rgba(0,0,0,0.2);

text-align: center;

width: 350px;

}

h2 {

color: #333;

margin-bottom: 20px;

}

input {

width: 90%;

padding: 10px;

margin-top: 10px;

border: 2px solid #ccc;

border-radius: 8px;



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font-size: 16px;

}

button {

margin-top: 15px;

padding: 10px 25px;

background: hsl(32, 39%, 49%);

color: white;

border: none;

border-radius: 8px;

font-size: 16px;

cursor: pointer;

transition: 0.3s;

}

button:hover {

background: hsl(60, 100%, 41%);

}

#result {

margin-top: 20px;

font-size: 20px;



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font-weight: bold;

color: #333;

}

</style>

</head>

<body>

<div class="container">

<h2>🎓 Sharvari's Scholarship Eligibility System [24070521207] </h2>

<label>Enter Marks:</label>

<input type="number" id="marks" placeholder="Enter marks (0-100)">

<button onclick="checkGrade()">Submit</button>

<p id="result"></p>

</div>

<script>

function checkGrade() {

var marks = Number(document.getElementById("marks").value);



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```
var input = document.getElementById("marks").value;

if (input === "") {

    alert("Please enter marks!");

    return;

}

if (marks < 0 || marks > 100) {

    alert("Please enter valid marks between 0 and 100.");

    return;

}

// Declared the scholarship variable properly

var scholarship = "";

if (marks >= 95) {

    scholarship = "Platinum Scholarship";

} else if (marks >= 85) {

    scholarship = "Gold Scholarship";
```



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```
} else if (marks >= 70) {  
  
    scholarship = "Silver Scholarship";  
  
} else if (marks >= 50) {  
  
    scholarship = "Bronze Scholarship";  
  
} else {  
  
    // Simplified the final fallback case  
  
    scholarship = "Not Eligible for Scholarship";  
  
}  
  
// Now variables match perfectly  
  
document.getElementById("result").innerHTML = "Marks: " + marks +  
"<br>Scholarship: " + scholarship;  
  
}  
  
</script>  
  
</body>  
  
</html>
```

```
marks = Number(marks);
if(marks < 0 || marks > 100)
{
    alert("Enter Valid Marks for Eligibility");
    return;
}
let grade;
if(marks >= 95)
    grade = "Platinum Scholarship";
else if(marks >= 85)
    grade = "Gold Scholarship";
else if(marks >= 70)
    grade = "Silver Scholarship";
else if(marks >=50)
    grade = "Bronze Scholarship";
else if(marks <= 50)
    grade = "Not Eligible for Scholarship";
document.getElementById("result").innerHTML =
    "Marks: " + marks + "<br>Grade: " + grade;
}
</script>
</body>
</html>
```



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3. Output:

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

Sharvari's Scholarship Eligibility System [24070521207]

Enter Marks:

Submit

Marks: 90
Scholarship: Gold Scholarship

C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

This page says

Please enter valid marks between 0 and 100.

OK

Sharvari's Scholarship Eligibility System [24070521207]



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C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac3.html

This page says

Please enter marks!

OK

 **Sharvari's Scholarship Eligibility System**
[24070521207]

Enter Marks:

Result/Conclusion: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web developmen



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● Experiment No.: 4

1. Title: Use functions types, scope, and closures; apply try-catch; build palindrome checker.

2. Software/Tools Required: VS Code, web browser, HTML, JavaScript.

3. Theory:

1. Regular Function

A regular function is a reusable block of code that performs a specific task. It can accept parameters, return values, and is invoked whenever needed. Functions improve code reusability and organization.

2. Function Declaration

A function declaration defines a function using the function keyword and a function name. It is hoisted, meaning it can be called before its declaration in the code.

3. Function Expression

A function expression stores a function inside a variable. Unlike function declarations, it is not fully hoisted and can only be called after it is defined.

4. Anonymous Function

An anonymous function is a function without a name. It is commonly used as a callback function or assigned to a variable for one-time or temporary operations.

5. Arrow Function

An arrow function is a shorter syntax for writing functions using the => operator. It is concise, easier to read, and automatically inherits the this value from its surrounding scope.

6. Variable Scope

Variable scope determines where a variable can be accessed in a program. JavaScript provides global scope, function scope, and block scope (using let and const).



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4. Output :

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac4Palindrome.html

 **Palindrome Checker**

Check Now

✓ **"2442" is a Palindrome**

 Checks Performed : 1

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac4Palindrome.html

 **Palindrome Checker**

Check Now

✗ **"2449" is NOT a Palindrome**

 Checks Performed : 2

5. Result/Conclusion: The JavaScript concepts such as functions, scope, closures, arrow functions, and exception handling were successfully studied and implemented. Their behaviour and usage were verified through simple programs. JavaScript functions and scope mechanisms help in writing clean, reusable, and efficient code. Understanding these concepts improves programming skills and enables the development of reliable and maintainable web app



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Case Study No.: 4

1. **Case Study Title: ATM system verifies if pin is palindrome. if palindrome,mesplay security message . to implement this create a function to reverse the entered pin number and another function to check if it is palindrome. use function declaraction,function expression ,arrow function and demonstrate scope and closure concepts.**

Case Study Program Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>ATM PIN Verification</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial, sans-serif;
```

```
background-color: #f5f0eb; /* Soft beige background */
```

```
color: #4a3c31; /* Deep dark brown text */
```

```
display: flex;
```

```
flex-direction: column;
```

```
align-items: center;
```

```
justify-content: center;
```



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height: 100vh;

margin: 0

}

.card {

background-color: #ffffff;

padding: 30px;

border-radius: 10px;

box-shadow: 0 4px 10px rgba(74, 60, 49, 0.15);

text-align: center;

border-top: 5px solid #6b4c3b; /* Medium warm brown accent */

}

h2 {

color: #5c3a21; /* Rich chocolate brown */

}

label {

display: block;

margin-bottom: 10px;

font-weight: bold;

}



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/* Changed selector from input[type="password"] to input[type="text"] */

input[type="text"] {

padding: 8px;

width: 200px;

border: 1px solid #b8a398;

border-radius: 4px;

margin-bottom: 15px;

}

button {

background-color: #6b4c3b; /* Warm brown button */

color: white;

padding: 9px 18px;

border: none;

border-radius: 4px;

cursor: pointer;

font-weight: bold;

}

button:hover {

background-color: #5c3a21;



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}

#result {

margin-top: 20px;

font-size: 14px;

}

</style>

</head>

<body>

<div class="card">

<h2>Sharvari's (1207) PIN Verification</h2>

<label for="pin">Enter ATM PIN:</label>

<input type="text" id="pin">

<button onclick="verifyPIN()">Verify</button>

<h3 id="result"></h3>

</div>

<script>



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// Function Declaration - defines a function with the function keyword and a function name (use when main part of program and you want it available thruout the scope)

```
function reversePIN(pin) {  
  
    return pin.split("").reverse().join("");  
  
}
```

// Function Expression - stores a function inside the variable (use when passing functions as arguments, assigning functions to variables , creating callbacks)

```
const isPalindrome = function(pin) {  
  
    return pin === reversePIN(pin);  
  
};
```

// Arrow Function - shorter way to write a function in js. it was introduced in ES6 and uses arrow syntax

```
const displayMessage = (msg) => {  
  
    document.getElementById("result").innerHTML = msg;  
  
};
```

// Scope and Closure

```
function verifyPIN() {
```



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```
let pin = document.getElementById("pin").value;

function securityMessage() {

    if (isPalindrome(pin)) {

        return " PIN is a Palindrome. Hence, Its a weak password. Please Change it for
max security";

    } else {

        return " PIN is Not a Palindrome. So, security verified!";

    }

}

displayMessage(securityMessage());

}

</script>

</body>

</html>
```



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Output:



Sharvari's (1207) PIN Verification

Enter ATM PIN:

PIN is a Palindrome. Hence, Its a weak password. Please Change it for max security



Sharvari's (1207) PIN Verification

Enter ATM PIN:

PIN is Not a Palindrome. So, security verified!



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3. Result/Conclusion: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



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Case Study No.: 5

1. Case Study Title: Implementation of JavaScript Array Operations Using Built-in Methods

Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">

  <title>Array Operations,Min/Max, Slice/Splice
[24070521207 SharvariWakde]</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f2f2f2;
      text-align: center;
      padding: 40px;
    }

    .container {
      width: 500px;
      margin: auto;
      background: white;
      padding: 30px;
      border-radius: 12px;
      box-shadow: 0 0 10px gray;
    }

    input, select {
      width: 90%;
      padding: 10px;
      margin: 8px;
      font-size: 16px;
    }

    button {
      padding: 12px 25px;
      margin-top: 10px;
      background: #333;
      color: white;
      border: none;
      border-radius: 5px;
```



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```
<input type="number" id="sliceStart">

<label>Slice End:</label>
<input type="number" id="sliceEnd">
<br><br>
<button onclick="arrayOperations()">Perform
Operations</button>
<p id="output"></p>
<script src="shreya_1196_case5.js"></script>
</body>
</html>
```

```
function arrayOperations() {
    let arr = document.getElementById("arrayInput").value
        .split(",") .map(Number);
    let pushValue =
Number(document.getElementById("pushValue").value);
    let unshiftValue =
Number(document.getElementById("unshiftValue").value);

    let spliceIndex =
Number(document.getElementById("spliceIndex").value);
    let spliceValue =
Number(document.getElementById("spliceValue").value);

    let sliceStart =
Number(document.getElementById("sliceStart").value);
    let sliceEnd =
Number(document.getElementById("sliceEnd").value);
    let output = "";
    output += "Original Array: " + arr + "<br><br>";
    arr.push(pushValue);
    output += "After push: " + arr + "<br>";
```



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```
// pop()
arr.pop();
output += "After pop: " + arr + "<br>";

// unshift()
arr.unshift(unshiftValue);
output += "After unshift: " + arr + "<br>";

// shift()
arr.shift();
output += "After shift: " + arr + "<br>";

// splice()
arr.splice(spliceIndex, 1, spliceValue);
output += "After splice: " + arr + "<br>";

// slice()
let newArray = arr.slice(sliceStart, sliceEnd);
output += "After slice: " + newArray + "<br><br>";

// Maximum and Minimum
let maximum = Math.max(...arr);
let minimum = Math.min(...arr);

output += "Maximum: " + maximum + "<br>";
output += "Minimum: " + minimum;

document.getElementById("output").innerHTML = output;
}
```



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3. Output:

Array Operations,Min/Max, Slice

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/casestudy5.html

Array Operations, Min/Max, Splice/slice [24070521207_Sharvariwakde]

Enter numbers separated by commas:

What do you want to do?

Find Maximum

Submit

Maximum

Array: 10, 20, 30, 40, 50

Maximum value: 50

4. Result/Conclusion:

Thus, various JavaScript array operations were successfully performed and the results were displayed using user input.



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Case Study No.: 8

- Case Study Title:** Online Job Application Form, A company wants to create an online job application form that checks the entered information immediately and prevents submission when the details are invalid.

Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Job Application Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: linear-gradient(135deg, #dbeafe, #e0e7ff);
      margin: 0;
      padding: 25px;
    }
    .container {
      width: 650px;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 15px;
      box-shadow: 0 5px 20px rgba(0,0,0,0.15);
    }
    h2 {
      text-align: center;
      color: #3730a3;
    }
    h3 {
      color: #4f46e5;
      border-bottom: 2px solid #ddd;
      padding-bottom: 5px;
      margin-top: 25px;
    }
  </style>
</head>
<body>
  <div class="container">
    <h2>Online Job Application Form</h2>
    <h3>Personal Information</h3>
    <div>
      <input type="text" value="Name" />
      <input type="text" value="Email" />
      <input type="text" value="Phone" />
    </div>
    <input type="button" value="Submit" />
  </div>
</body>
</html>
```



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```
label {
    display: block;
    margin-top: 12px;
    font-weight: bold;
}

input, select, textarea {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    border: 1px solid #ccc;
    border-radius: 6px;
    box-sizing: border-box;
}

textarea {
    height: 70px;
    resize: none;
}

.radio {
    width: auto;
    margin-right: 5px;
}

.message {
    font-size: 13px;
    margin-top: 4px;
}

.error {
    color: red;
}

.success {
    color: green;
}

button {
    width: 100%;
    padding: 12px;
    margin-top: 20px;
    background-color: #4f46e5;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    cursor: pointer;
```



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```
}

button:hover {

    background-color: #3730a3;

}

#result {

    text-align: center;

    margin-top: 20px;

    font-weight: bold;

}

</style>
</head>
<body>
<div class="container">

    <h2>📁 Shreya's 1196 Online Job Application Form</h2>
    <form id="jobForm">

        <h3>Personal Information</h3>

        <!-- NAME -->

        <label>Full Name:</label>

        <input type="text" id="name" placeholder="Enter full name">

        <div id="nameMessage" class="message"></div>

        <!-- DOB -->

        <label>Date of Birth:</label>

        <input type="date" id="dob">

        <div id="dobMessage" class="message"></div>

        <!-- GENDER -->

        <label>Gender:</label>

        <input type="radio" class="radio" name="gender" value="Male">
Male

        <input type="radio" class="radio" name="gender" value="Female">
Female

        <input type="radio" class="radio" name="gender" value="Other">
Other

        <div id="genderMessage" class="message"></div>

        <!-- EMAIL -->

        <label>Email Address:</label>

        <input type="text" id="email" placeholder="Enter email">

        <div id="emailMessage" class="message"></div>

        <!-- MOBILE -->

        <label>Mobile Number:</label>

        <input type="text" id="mobile"

            placeholder="Enter 10-digit mobile number"

    </div>

</body>
</html>
```



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```
maxlength="10">

<div id="mobileMessage" class="message"></div>

<!-- ADDRESS -->
<label>Address:</label>
<textarea id="address" placeholder="Enter address"></textarea>
<div id="addressMessage" class="message"></div>

<!-- CITY -->
<label>City:</label>
<input type="text" id="city" placeholder="Enter city">
<div id="cityMessage" class="message"></div>

<!-- STATE -->
<label>State:</label>
<input type="text" id="state" placeholder="Enter state">
<div id="stateMessage" class="message"></div>

<!-- PIN -->
<label>PIN Code:</label>
<input type="text" id="pin"
      placeholder="Enter 6-digit PIN"
      maxlength="6">
<div id="pinMessage" class="message"></div>

<!-- QUALIFICATION -->
<label>Educational Qualification:</label>
<select id="qualification">
  <option value="">Select qualification</option>
  <option>B.Tech</option>
  <option>B.E.</option>
  <option>BCA</option>
  <option>MCA</option>
  <option>M.Tech</option>
  <option>Other</option>
</select>
<div id="qualificationMessage" class="message"></div>

<!-- PERCENTAGE -->
<label>Percentage:</label>
<input type="number" id="percentage"
      placeholder="Enter percentage">
<div id="percentageMessage" class="message"></div>

<!-- PASSING YEAR -->
<label>Passing Year:</label>
<input type="number" id="passingYear"
      placeholder="Enter passing year">
```



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```
<div id="yearMessage" class="message"></div>

<h3>Professional Information</h3>

<!-- POSITION -->
<label>Applying for Job Position:</label>
<input type="text" id="position"
        placeholder="Example: Software Developer">
<div id="positionMessage" class="message"></div>

<!-- EXPERIENCE -->
<label>Total Work Experience (Years):</label>
<input type="number" id="experience"
        placeholder="Enter experience">
<div id="experienceMessage" class="message"></div>

<!-- ORGANIZATION -->
<label>Current / Previous Organization:</label>
<input type="text" id="organization"
        placeholder="Enter organization">
<div id="organizationMessage" class="message"></div>

<!-- DESIGNATION -->
<label>Current Designation:</label>
<input type="text" id="designation"
        placeholder="Example: Intern">
<div id="designationMessage" class="message"></div>

<!-- SALARY -->
<label>Expected Salary:</label>
<input type="number" id="salary"
        placeholder="Enter expected salary">
<div id="salaryMessage" class="message"></div>

<!-- SKILLS -->
<label>Technical Skills:</label>
<textarea id="skills"
        placeholder="Example: Java, Python, SQL,
HTML"></textarea>
<div id="skillsMessage" class="message"></div>
<button type="submit">Submit Application</button>
</form>

<div id="result"></div>
</div>

<script>
    let form = document.getElementById("jobForm");
    // FUNCTION TO SHOW MESSAGE
    function showMessage(id, message, type) {
```



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```
document.getElementById(id).className = "message " + type;
document.getElementById(id).innerHTML = message;

// NAME
document.getElementById("name").addEventListener("input", function()
{
    if (this.value.trim() === "") {
        showMessage("nameMessage", "Name is required", "error");
    } else {
        showMessage("nameMessage", "✓ Valid name", "success");
    }
});

// DATE OF BIRTH
document.getElementById("dob").addEventListener("change", function()
{
    if (this.value === "") {
        showMessage("dobMessage", "Date of birth is required",
"error");
    } else {
        showMessage("dobMessage", "✓ Date of birth selected",
"success");
    }
});

// GENDER
document.querySelectorAll('input[name="gender"]').forEach(function(radi
o) {
    radio.addEventListener("change", function() {
        showMessage("genderMessage", "✓ Gender selected",
"success");
    });
});

// EMAIL
document.getElementById("email").addEventListener("input",
function() {
    let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (pattern.test(this.value)) {
        showMessage("emailMessage", "✓ Valid email", "success");
    } else {
        showMessage("emailMessage", "Invalid email address",
"error");
    }
});
```



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```

    });
    // MOBILE
    document.getElementById("mobile").addEventListener("input",
function() {
    this.value = this.value.replace(/^[^0-9]/g, "");
    if (this.value.length === 10) {
        showMessage("mobileMessage", "✓ Valid mobile number",
"success");
    } else {
        showMessage("mobileMessage", "Invalid mobile number - enter
10 digits", "error");
    }
});
    // ADDRESS
    document.getElementById("address").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("addressMessage", "Address is required",
"error");
    } else {
        showMessage("addressMessage", "✓ Valid address",
"success");
    }
});
    // CITY
    document.getElementById("city").addEventListener("input", function()
{
    if (this.value.trim() === "") {
        showMessage("cityMessage", "City is required", "error");
    } else {
        showMessage("cityMessage", "✓ Valid city", "success");
    }
});
    // STATE
    document.getElementById("state").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("stateMessage", "State is required", "error");
    } else {
        showMessage("stateMessage", "✓ Valid state", "success");
    }
});

```



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```
}

});

// PIN
document.getElementById("pin").addEventListener("input", function()
{
    this.value = this.value.replace(/^[^0-9]/g, "");
    if (this.value.length === 6) {
        showMessage("pinMessage", "✓ Valid PIN code", "success");
    } else {
        showMessage("pinMessage", "Invalid PIN - enter 6 digits",
"error");
    }
});

// QUALIFICATION
document.getElementById("qualification").addEventListener("change",
function() {
    if (this.value === "") {
        showMessage("qualificationMessage",
            "Please select qualification", "error");
    } else {
        showMessage("qualificationMessage",
            "✓ Qualification selected", "success");
    }
});

// PERCENTAGE
document.getElementById("percentage").addEventListener("input",
function() {
    let value = Number(this.value);
    if (value >= 0 && value <= 100) {
        showMessage("percentageMessage",
            "✓ Valid percentage", "success");
    } else {
        showMessage("percentageMessage",
            "Invalid percentage - enter 0 to 100", "error");
    }
});

// PASSING YEAR
document.getElementById("passingYear").addEventListener("input",
function() {
    let year = Number(this.value);
    let currentYear = new Date().getFullYear();
```



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```
if (year >= 1950 && year <= currentYear) {
    showMessage("yearMessage",
        "✓ Valid passing year", "success");
} else {
    showMessage("yearMessage",
        "Invalid passing year", "error");
}
});
// JOB POSITION
document.getElementById("position").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("positionMessage",
            "Job position is required", "error");
    } else {
        showMessage("positionMessage",
            "✓ Valid job position", "success");
    }
});
// EXPERIENCE
document.getElementById("experience").addEventListener("input",
function() {
    let value = Number(this.value);

    if (value >= 0) {
        showMessage("experienceMessage",
            "✓ Valid experience", "success");
    } else {
        showMessage("experienceMessage",
            "Experience cannot be negative", "error");
    }
});
// ORGANIZATION
document.getElementById("organization").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("organizationMessage",
            "Organization is required", "error");
    } else {
        showMessage("organizationMessage",
```



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```
        "✓ Valid organization", "success");

    }

});

// DESIGNATION
document.getElementById("designation").addEventListener("input",
function() {
    if (this.value.trim() === "") {
        showMessage("designationMessage",
            "Designation is required", "error");
    } else {
        showMessage("designationMessage",
            "✓ Valid designation", "success");
    }
});

// SALARY
document.getElementById("salary").addEventListener("input",
function() {
    if (Number(this.value) > 0) {
        showMessage("salaryMessage",
            "✓ Valid expected salary", "success");
    } else {
        showMessage("salaryMessage",
            "Enter a valid salary", "error");
    }
});

// TECHNICAL SKILLS
document.getElementById("skills").addEventListener("input",
function() {

    if (this.value.trim() === "") {
        showMessage("skillsMessage",
            "Technical skills are required", "error");
    } else {
        showMessage("skillsMessage",
            "✓ Skills entered", "success");
    }
});
```



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```
// FORM SUBMISSION

form.addEventListener("submit", function(event) {

    event.preventDefault();

    let name = document.getElementById("name").value.trim();
    let dob = document.getElementById("dob").value;
    let gender =
document.querySelector('input[name="gender"]:checked');
    let email = document.getElementById("email").value;
    let mobile = document.getElementById("mobile").value;
    let address = document.getElementById("address").value.trim();
    let city = document.getElementById("city").value.trim();
    let state = document.getElementById("state").value.trim();
    let pin = document.getElementById("pin").value;
    let qualification =
document.getElementById("qualification").value;
    let percentage =
Number(document.getElementById("percentage").value);
    let passingYear =
Number(document.getElementById("passingYear").value);

    let position = document.getElementById("position").value.trim();
    let experience =
Number(document.getElementById("experience").value);
    let organization =
document.getElementById("organization").value.trim();
    let designation =
document.getElementById("designation").value.trim();
    let salary = Number(document.getElementById("salary").value);
    let skills = document.getElementById("skills").value.trim();

    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    let currentYear = new Date().getFullYear();
    if (
        name !== "" &&
        dob !== "" &&
        gender &&
        emailPattern.test(email) &&
        mobile.length === 10 &&
        address !== "" &&
        city !== "" &&
        state !== "" &&
```



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```
pin.length === 6 &&
qualification !== "" &&
percentage >= 0 &&
percentage <= 100 &&
passingYear >= 1950 &&
passingYear <= currentYear &&
position !== "" &&
experience >= 0 &&
organization !== "" &&
designation !== "" &&
salary > 0 &&
skills !== ""
) {
    document.getElementById("result").className = "success";

    document.getElementById("result").innerHTML =
        "✅ Application Submitted Successfully!";

    form.reset();

} else {

    document.getElementById("result").className = "error";
    document.getElementById("result").innerHTML =
        "❌ Please correct the invalid details before
submitting.";
    }
});
</script>
</body>
</html>
```

3. Output:



॥ वसुधैव कुटुम्बकम् ॥

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Online Job Application

C:/Users/HP/Desktop/24070521207_SharvariW(S)/casestudy&html

Sharvari's 1207 Job Application Form

Apply For Job.

Personal Information

Full Name *

sharvari

✓ Valid name

Date of Birth *

01 - 12 - 2001

✓ Date of birth selected

Gender *

☐ Male

☒ Female

☐ Other

✓ Gender selected

Email Address *

sharvari@gmail.com

Invalid email address

Mobile Number *

786544321

Invalid mobile - enter 10 digits

PIN Code *

440019

✓ Valid PIN code

Address *

nagpur, maharashtra

✓ Valid address

City *

nagpur

✓ Valid city

State *

maharashtra

✓ Valid state

Educational Information

Educational Qualification *

B.Tech

✓ Qualification selected

Percentage *

99

✓ Valid percentage

Passing Year *

1800

Invalid passing year

Professional Information

Applying for Job Position *

MI engineer

✓ Valid job position

Total Work Experience (Years) *

2

✓ Valid experience

Current / Previous Organization *

Enter organization

Current Designation *

Example: Intern

Mouza - Wathoda, Nagpur 440008, Maharashtra, India. | Tel: +91-712-6192370 / +91-712-6192368
Email: info@sitnagpur.siu.edu.in | Website: www.sitnagpur.edu.in

4. Result/Conclusion:

Thus, an online job application form was successfully created with **live validation** to check user details and prevent invalid form submission.



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● Experiment No.:

1. **Title:** Implement localStorage and sessionStorage to save user preferences like theme settings.
2. **Software/Tools Required:** VS Code, JavaScript,html.
3. **Theory:**

Web Storage is used to store data in the browser using key-value pairs. In this program, localStorage is used to store data permanently, while sessionStorage stores data only for the current browser session. The functions `setItem()`, `getItem()`, and `removeItem()` are used to save, retrieve, and delete the stored preferences. The program allows the user to select Light, Dark, or Blue themes.

The selected theme can be changed dynamically using `classList.add()` and `classList.remove()`. `getElementById()` is used to access the required HTML elements, while `innerText` is used to display the current theme, storage type, saved values, and messages. The `loadPreference()` function loads saved settings when the page opens, and `savePreference()` stores the selected settings.

The `removePreference()` function clears the saved preferences and resets the theme to Light. Overall, this program demonstrates how Web Storage and DOM manipulation can be used together to save user preferences and change.

The use of Web Storage makes the webpage more user-friendly because the selected preferences can be remembered without needing to enter them again. When the user selects a theme and storage type, the information is stored in the browser. On opening the webpage again, the saved information can be retrieved using `getItem()` and applied automatically. This helps in understanding how websites can remember basic user settings.

The program also shows the use of JavaScript for interacting with HTML and CSS elements. By using DOM methods and JavaScript functions, the webpage can respond to user actions and update its content without reloading the page.



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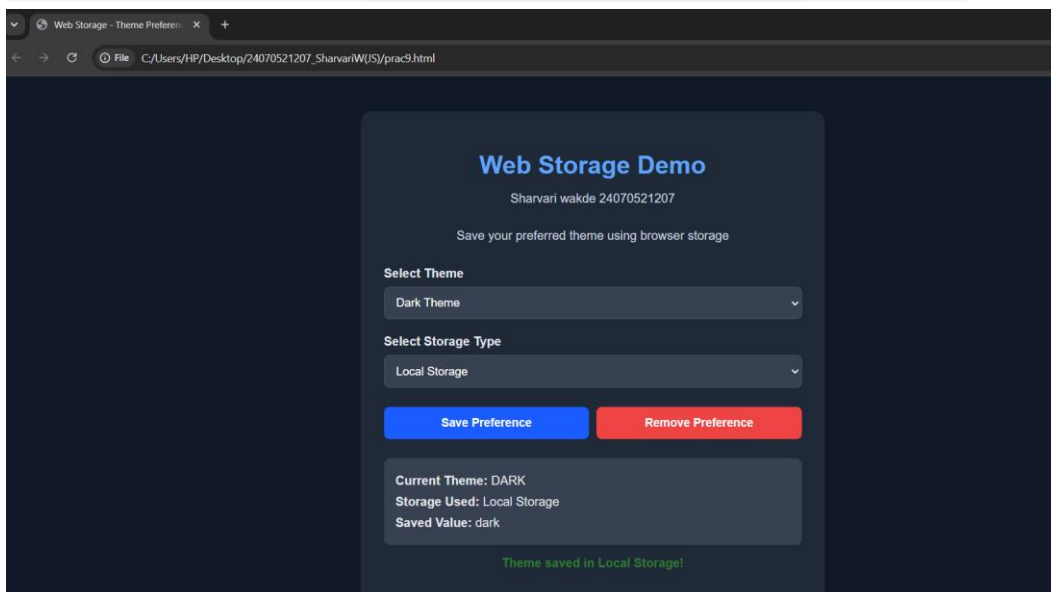
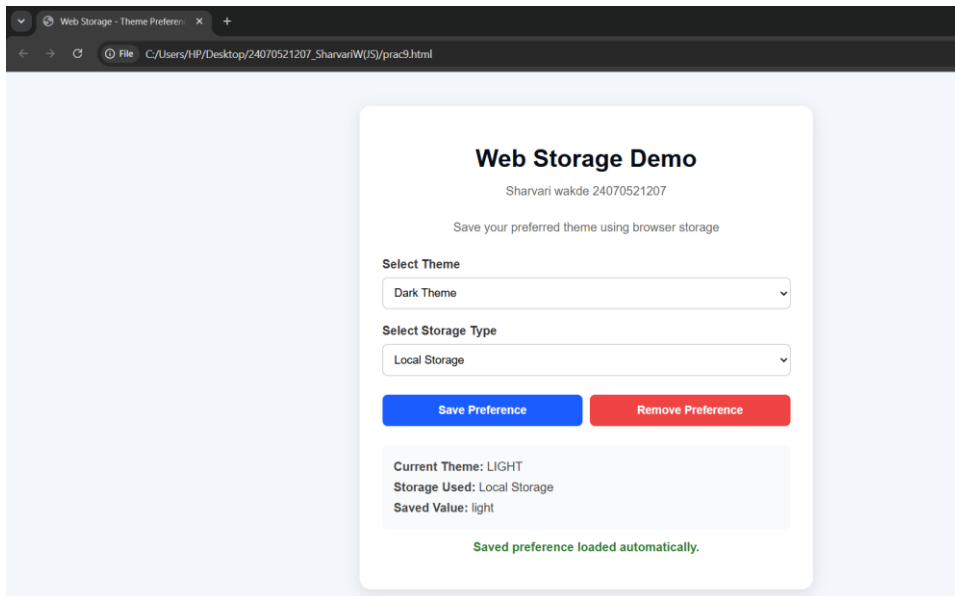
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4. Output :



5. Result/Conclusion:

Thus, local Storage and session Storage were successfully implemented to store and retrieve user theme preferences. The selected theme was applied dynamically, displayed on the webpage, automatically loaded when available, and removed successfully using JavaScript.



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● Experiment No.:

1. Title:

Load and display JSON data using `fetch()` and `jQuery $.getJSON()` ; render results dynamically in a table.

2. Software/Tools:

VS Code, JavaScript, HTML

3. Theory:

In this program, JSON data is used to store student information such as ID, name, and PRN. JavaScript is used to load and display this data on the webpage. The `fetch()` method is used to retrieve the JSON data asynchronously, and `response.json()` converts the response into JavaScript data so that it can be easily used in the program. jQuery also provides the `$.getJSON()` method, which is another simple way to load JSON data.

The `forEach()` function is used to go through each student record one by one. The `createElement()` method is used to create table rows dynamically, while `innerHTML` is used to add the student ID, name, and PRN into the table cells. `appendChild()` is then used to add the newly created rows to the table.

Overall, this program shows how JSON data can be fetched and displayed dynamically using JavaScript and jQuery. Instead of manually entering every student record in HTML, the records are automatically added to the table using JavaScript. This makes the webpage easier to update and manage.

The program also helps in understanding how JavaScript can work with external data and update the webpage dynamically. When the JSON data is loaded, the required information is taken from each object and displayed in the table. This makes the program more flexible because new student records can be added to the JSON data without changing the HTML table manually.



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4. Output :

The screenshot shows a web browser window titled 'JSON Data Display' with the address bar displaying 'File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac10.html'. The main content area has a light blue background and features a white card with the title 'sharvari 1207 JSON Data' in green. Below the title are two green buttons: 'Load using fetch()' and 'Load using \$.getJSON()'. A table with three columns (ID, Name, PRN) and three rows of student data is displayed below the buttons.

ID	Name	PRN
1	Sharvari	24070521207
2	Shreya	24070521196
3	Vithika	24070521192

This screenshot is identical to the one above, showing the same web browser window and content. It displays the 'sharvari 1207 JSON Data' card with the 'Load using fetch()' and 'Load using \$.getJSON()' buttons, and the table of student data.

ID	Name	PRN
1	Sharvari	24070521207
2	Shreya	24070521196
3	Vithika	24070521192

5. Result/Conclusion:

Thus, JSON student data containing ID, Name, and PRN was successfully loaded using both `fetch()` and jQuery `$.getJSON()` and dynamically displayed in an HTML table.